One daily link row. Assignment totals preserved.

A new postprocessing export combines each directed link's hourly speeds and approximate hourly volumes across the actual modeled periods. S3 supplies the native-interval volume pattern; one multiplier per link and period preserves assignment totals. Hourly aggregation follows this correction.

Regional links	Hourly speeds	Hourly volumes	Maximum total error
33,350	18	18	5.09e-11 vehicles

Step-by-step implementation

1. Resolve outputs and scopes. Run after both combined performance files are created in the resolved DTALite folder. Use exactly their jurisdiction and regional roadway scopes.

2. Identify the same directed link. Join converted network attributes by from_node_id + to_node_id. Export the network link_id, keeping the assignment ID separately in the audit.

3. Read actual period bounds. Use each period's settings.csv start/end hours; configured ranges are fallback when settings is absent. Build hourly bins from the first period start to the last period end.

4. Resolve link-level S3 inputs. Read known free speed, capacity and lanes. Derive critical speed or m from one another; apply a documented prior only where the information is missing.

5. Convert and conserve. Calculate whole-link flow, multiply by interval hours, sum within each period, then apply its assignment-total ratio to every interval.

6. Aggregate and write. Sum the corrected native interval counts into hourly bins and calculate duration-weighted harmonic speeds. Write exactly two feature CSVs, one per scope. Diagnostics are returned in memory for validation.

Implementation location

Branch: **codex/s3-speed-volume-export**
Calculation: postprocessing/pipeline/speed_volume.py
Automatic trigger: postprocessing/runner.py
Regression tests: tests/test_speed_volume.py
Reproduction: scripts/validate_speed_volume.py

Input files, columns and units

Inputs are the existing period link.csv and link_performance.csv files. The processed combined DataFrame supplies period-owned performance values; the exporter rereads converted link.csv for network attributes. Positive finite values are selected row by row.

Quantity	Input columns / precedence	Interpretation
Identity	Both files: from_node_id, to_node_id	Normalized directed pair. Opposite directions remain distinct; duplicate/unmatched pairs fail.
Output ID / type	link.csv: link_id, link_type	First configured period containing that link. Internal performance IDs stay in the audit.
Assigned total V	link_performance.csv: volume	Whole-link assignment count for that period. Never divide by lanes.
Incremental speeds	spd_mph_HH:MM; legacy spd_HH:MM or spd_kmph_HH:MM	Use mph; convert km/h with 1.609344. Prefer explicit mph for duplicate representations.
Free speed vf	link.csv:vdf_free_speed_mph; performance:free_speed_mph; link.csv:free_speed	First two are mph. Generic link.csv free_speed is km/h and is divided by 1.609344.
Lane count	link.csv:lanes	Directed lane count for the current period. Assume 1 only when unavailable/invalid; audit this.
Capacity C	performance:link_capacity; link.csv:capacity; performance:lane_capacity	link_capacity is already veh/hour/link. Other two are veh/hour/lane; multiply by current lanes once.
Critical speed vc	speed_at_capacity_mph, vc_mph, fd_vc_mph	Explicit values first. Otherwise derive from m; final fallback is 0.75 vf, matching the kernel default.
Shape m	s3_shape_m, s3_m	Performance first, then network. Derive missing m from vf and vc.

Whole-link example

A three-lane link with capacity 1,800 veh/hour/lane has C = 5,400 veh/hour. If link_capacity already equals 5,400, use 5,400 directly. The resulting density, flow and interval volumes all describe the whole directed link.

The requested fallback ratio now matches the C++ kernel: vc = 0.75 vf. Explicit S3 inputs retain their existing priority. QVDF vdf_s, vdf_n and boundary speed anchors are not interpreted as S3 shape parameters.

S3 inversion and period scaling

Derive parameters before assuming them

If vf and m are known, derive vc. If vf and vc are known, derive m. When vc and m conflict, supplied valid vc is authoritative and m is recomputed. If vf is missing but vc and m exist, derive vf first.

```
vc = vf / 2^(2/m)
m = 2 ln(2) / ln(vf/vc)
kc = C / vc
```

When neither critical speed nor shape is supplied, use vc = 0.75 vf, giving m = 4.819, consistent with the kernel's default ratio. Known network vf is preserved. Missing vf falls back to speed p95 clipped to 35-85 mph; absent data uses 35 mph. Missing C uses the CBI batch prior of 1,800 veh/hour/lane times lanes.

Invert each interval's speed

```
v* = min(v, 0.99 vf)
r = max(vf / max(v*, 1 mph), 1.00001)
k = kc [max(r^(m/2) - 1, 1e-8)]^(1/m)
q = min(v* k, C) [veh/hour/link]
u_i = q_i dt_i_hours [vehicles/link]
```

These numerical safeguards follow CBI's inverse S3 function. The code evaluates density in log form to avoid exponent overflow. The saved supplied speeds are preserved; the 0.99 vf limit applies inside inversion.

One common ratio for all intervals of one period

```
U = sum(u_i over the period)
R = assigned period volume V / U
approx_volume_i = R u_i
sum(approx_volume_i) = V
```

Each link/period gets its own R. After this correction, sum interval counts into hourly bins. Hourly speed = sum(dt) / sum(dt / speed), using duration weights. For equal five-minute samples this is n / sum(1/speed). Floating-point residue is corrected before aggregation; counts are not rounded to integers.

Illustrative five-minute bin	Raw interval count	R	Saved count
1	100	2.0	200
2	200	2.0	400
3	150	2.0	300
Period total	450	V / U = 900 / 450	900 = assignment

The CBI reconstruction module uses an additive capacity-constrained adjustment. This implementation uses the requested multiplicative adjustment. Final scaled flow may exceed C; the audit records it rather than clipping away assigned demand.

File schema and explicit assumptions

New file beside combined outputs	Contents
link_speed_volume.csv	Jurisdiction-scoped daily rows
link_speed_volume_regional.csv	Regional daily rows
In-memory diagnostics	Parameters, sources, totals, scale and flags. The validation script stores these in validation_diagnostics; production writes only the two CSVs above.

Daily column order

1. link_id, from_node_id, to_node_id, link_type, free_speed (mph), capacity (veh/hour/link), lanes.
2. All spd_mph_START_END columns in modeled order.
3. All approx_volume_START_END columns in the same order. Example: spd_mph_06:00_07:00 and approx_volume_06:00_07:00. Values are vehicles during that bin.

The seven static fields describe the first configured period containing a link. Each calculation uses the actual attributes for its own period; the audit retains those effective values. A link absent from a period retains blank cells for that period.

Cases handled explicitly

Case	Behavior / audit
Hourly speed / volume	Speed = sum(dt) / sum(dt/v), using duration weights rather than volume weights. Volume = sum of already-scaled native counts. A zero speed gives zero harmonic speed; invalid supplied speed makes the hourly speed blank.
Supplied invalid speed cells	Keep the original cell. Give it zero raw flow weight and record invalid_speed_intervals. Other valid intervals receive the allocation.
Positive assignment, zero raw S3 total	Allocate in proportion to interval duration. Status uniform_duration_fallback; scale factor is blank.
Zero assigned volume	Every interval count is zero.
Duplicate/unmatched pairs, invalid assigned volume, overlapping periods	Clear error; no silent attribution of another link's parameters.
Legacy file with no speed columns	Skip new exports with a warning. Clear stale same-named exports; existing combined/statistics behavior remains.

Actual modeled span, not an assumed calendar day

NT settings.csv confirms 19:00-24:00: conversion truncates the configured cross-midnight period at midnight. The new output therefore spans 06:00-24:00, with 18 hourly bins and no 00:00-06:00 extension. A 06:30 start would give 06:30-07:30 bins; the final bin is shortened if necessary. True overnight runs use increasing labels such as 24:00-25:00.

Validation on the true-shape network

Reprocessed copies of the existing DTALiteNetworkInput_TRUESHP_run outputs through the normal postprocessing entry point. This tests export/integration using completed assignment results; assignment itself was not rerun. All 105 automated project tests passed in a temporary checkout outside Dropbox.

Scope	Unique links	Columns	Hourly speed / volume bins
Jurisdiction	12,559	43	18 / 18
Regional	33,350	43	18 / 18

Every exported link/period total checked after reading the CSV

Scope	Period	Links	Bins	Assigned = exported*	Max error
Jurisdiction	AM	12,559	3	26,245,271.057	1.09e-11
Jurisdiction	MD	12,559	6	49,409,812.983	1.09e-11
Jurisdiction	PM	12,559	4	42,533,244.954	5.09e-11
Jurisdiction	NT	12,559	5	32,997,768.242	5.09e-11
Regional	AM	33,343	3	68,270,173.453	1.09e-11
Regional	MD	33,350	6	126,063,225.338	1.09e-11
Regional	PM	33,343	4	109,365,773.972	5.09e-11
Regional	NT	33,350	5	83,825,741.966	5.09e-11

*Equality within absolute 1e-8 and relative 1e-12 numerical tolerance. Error is the maximum per-link difference in vehicles after CSV round-trip, rather than only a network-wide comparison.

What the test confirmed

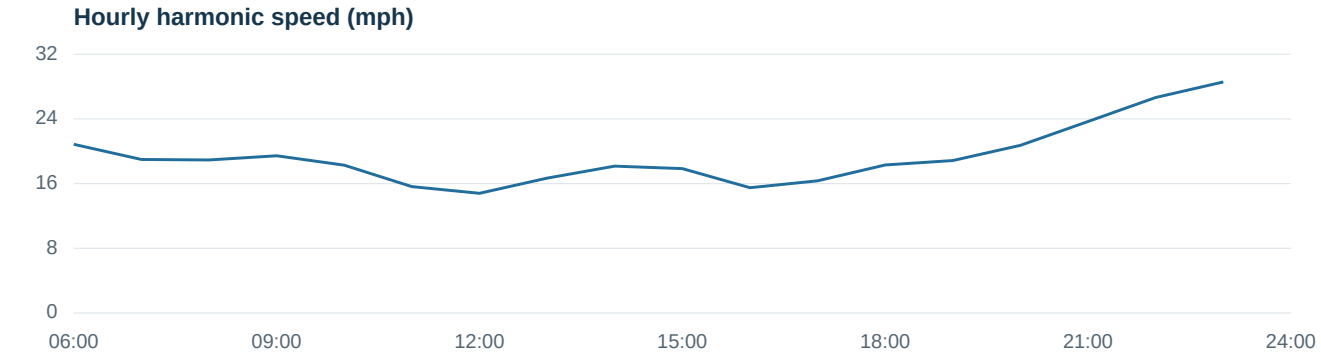
All 133,386 regional link/period records used their native link_capacity and network vdf_free_speed_mph. Critical speed was unavailable, so all used 0.75 vf and derived m. There were 0 invalid speed cells, 0 imputed cells and no duration-allocation fallbacks. Diagnostics record 328,281 native intervals above capacity after scaling.

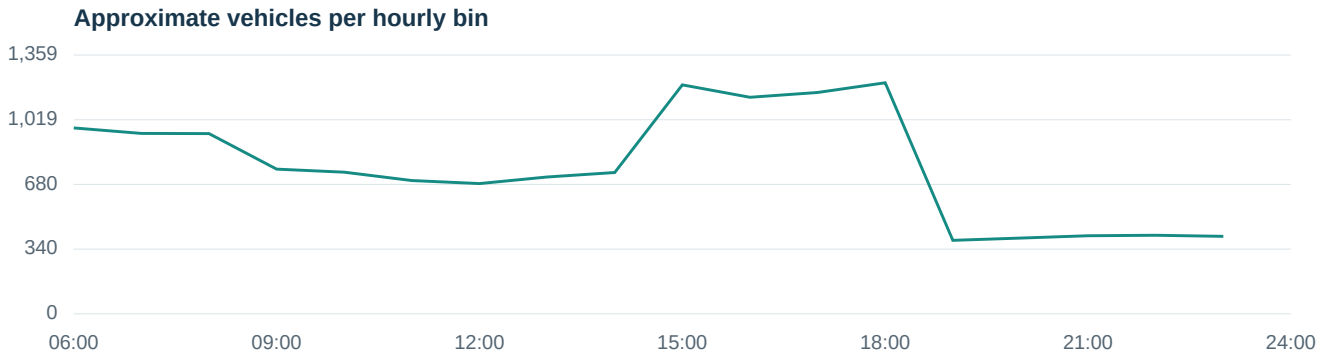
931 regional links changed capacity between periods, confirming why period-specific attributes must be used. The raw input SHA256 hashes were unchanged: **True**.

Regression coverage includes S3 capacity recovery, link capacity, scaling before aggregation, harmonic speeds, shifted hour origins, partial final hours, bins crossing period boundaries, settings fallback, 24:00/overnight ordering, zero/invalid speeds, duplicate pairs and exactly two production exports.

Example daily profile and reproducibility

Regional example: link 8250, directed pair 20000 to 20797. Hourly harmonic speeds and summed interval counts cover only the actual 06:00-24:00 span. Volume levels reflect independent period scale factors.





Period	Assigned volume	Raw S3 volume	Scale R	Exported total
AM	2,870.499	1,416.704	2.02618	2,870.499
MD	4,348.883	2,697.810	1.61200	4,348.883
PM	4,715.883	1,794.232	2.62836	4,715.883
NT	2,013.428	2,433.876	0.82725	2,013.428

Reproduce and inspect

Run scripts/validate_speed_volume.py with --source pointing to existing period outputs and --output to a new test folder. It stages copies, runs normal postprocessing, writes validation_summary.json and ten-row sample CSVs, and hashes source inputs before and after.

Outputs for this report: **nvta-run/s3-hourly-validation-20261105/**
Specification: **docs/s3-speed-volume.md**
All requested changes are on **codex/s3-speed-volume-export**. Publishing is a separate release step after review.

Reference: CBI fundamental_diagram.py (pinned source); s3_speed, inverse_s3_flow and synthetic_s3_context. The 1,800 veh/hour/lane prior is from CBI batch.py. No CBI dependency was added to the installed workflow.